The RESOLVE Swiss trial for people with chronic low back pain: statistical analysis plan for a cluster randomised controlled trial

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Abstract

Background. Statistical analysis plans describe the planned data management and analysis of a clinical trial before the data are seen, and so protect the eventual results from data-driven analytical choices. This paper reports the statistical analysis plan for the RESOLVE Swiss trial, which evaluates graded sensorimotor retraining combined with pain neuroscience education against usual physiotherapy for people with chronic low back pain.

Design and analysis. RESOLVE Swiss is a two-arm parallel-group *cluster* randomised controlled trial in Swiss physiotherapy practices. The practice is the unit of randomisation and the patient is the unit of analysis. The primary outcome is back-specific disability, measured with the Roland–Morris Disability Questionnaire at 18 weeks after the baseline measurement. The primary analysis is a linear mixed model over the baseline and 18-week measurements with fixed effects for time and the treatment-by-time interaction, no main effect of treatment, and random intercepts for practice and participant. The model constrains the two arms to a common baseline mean, which is how the baseline information enters, and the treatment effect is estimated with Kenward–Roger degrees of freedom, the small-sample correction that guards against anti-conservative mixed-model inference when the number of practices is small. A comparative Bayesian analysis with prespecified priors, informed by the individual participant data of the Australian trial, accompanies the frequentist results. We report the intra-cluster correlation with its uncertainty, the analyses of secondary outcomes, the handling of missing data, non-adherence and contamination, and the planned reporting.

Conclusion. Prespecifying the analysis reduces the scope for bias in the results of RESOLVE Swiss and makes explicit how the small number of clusters is dealt with.

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Keywords: Low back pain; Cluster randomised trial; Statistical analysis plan; Mixed models; Physical therapy.

1 Introduction

1.1 Background and rationale

Chronic low back pain is among the largest contributors to years lived with disability worldwide, and the effects of the treatments physiotherapists currently offer are, on average, modest. The Australian RESOLVE trial tested whether a treatment aimed at the central nervous system representation of the back (graded sensorimotor precision training preceded by pain neuroscience education) reduces pain more than a sham intervention. It reported a modest benefit in pain at 18 weeks, consistent with a clinically meaningful between-group difference, and a benefit in disability that persisted through 52 weeks [1]. An earlier and much smaller Swiss pilot compared a related programme with usual physiotherapy and pointed in the same direction without being able to resolve the question [2].

Two things are unresolved. First, RESOLVE compared the intervention with a sham. The clinically relevant comparison for Swiss practice is against usual physiotherapy as it is actually delivered. Second, the intervention has so far been delivered in a trial setting by trial therapists. RESOLVE Swiss therefore delivers it in ordinary physiotherapy practices, by the therapists who work there. That choice makes the trial pragmatic, and it forces the design to be a *cluster* randomised trial. People within a practice communicate directly. Therapists work side by side and share what they learn. Randomising patients within practices would therefore contaminate the control arm from the start. Communication between practices is possible too, but it is far less direct, so allocation at the level of the practice keeps contamination to a minimum [3, 4].

The trial's scientific advisory board recommended the cluster design for exactly this reason.

This analysis plan is written before the analysis of outcome data and follows the recommendations for the content of statistical analysis plans in clinical trials [5], together with the CONSORT extension for cluster randomised trials [4]. It reports the planned frequentist analysis of the primary and secondary outcomes, a comparative Bayesian analysis of the primary outcome, and a summary of the planned Bayesian interim analysis. [OPEN: cite the published study protocol once it is available]

2 Methods

2.1 Trial design

RESOLVE Swiss is a two-arm, parallel-group, cluster randomised, investigator blinded, multi-centre trial with 1:1 intended allocation at the level of the physiotherapy practice. Practices and institutions are the clusters. Patients treated within a practice are the units of observation and of analysis. Physical outcomes are assessed at baseline and at 18 weeks. All self-reported questionnaires, including the primary outcome, are collected again at 26 and 52 weeks after the baseline measurement.

The trial planned 20 physiotherapy institutions across Switzerland, and all 20 were randomised 1:1 (section 2.4). Since the allocation, six intervention practices and one control practice have withdrawn [OPEN: confirm numbers, timing and reasons], leaving 4 practices in the intervention arm and 9 in the control arm at the time of writing. The consequences of that split, and of any change to it, are quantified in section 2.6. The flow of practices, with reasons for withdrawal, has to be reported under the CONSORT cluster extension [4]. The imbalance of the withdrawals also matters in its own right (section 2.5).

The trial is a Category A study under the Swiss Ordinance on Clinical Trials (ClinO Art. 61). Ethical approval was granted by the Cantonal Ethics Committee of Zurich as lead committee (BASEC 2025-00784, decision of 20 January 2026, fulfilment of conditions confirmed on 10 April 2026). The trial is registered in the German Clinical Trials Register (DRKS00039237) and with Human Research Switzerland (HumRes67737).

2.2 Eligibility

Eligibility operates at two levels, the practice and the patient [4].

Practices. Swiss outpatient physiotherapy practices and institutions are eligible if they are willing to be allocated to either arm. Therapists in practices allocated to the intervention must complete a five-day RESOLVE training course and have their skills assessed before delivering any study treatment [OPEN: confirm any further practice-level criteria, e.g. minimum patient volume].

Participants. Eligible participants are adults aged 18 to 75 referred to physiotherapy for low back pain, with non-specific low back pain persisting for at least three months, a baseline RMDQ of at least 7 points, sufficient German to follow the study procedures and complete the questionnaires, internet access and willingness to use a private device, and a person available to assist with home training. Participants are excluded for evidence of specific spinal pathology (malignancy, fracture, infection, inflammatory joint or bone disease), radiculopathy, pregnancy or being less than six months postpartum, a coexisting major medical disease contraindicating general exercise, spinal surgery within the preceding 12 months, an intra-articular or perineural lumbar steroid injection within the preceding three months, inability to follow the study procedures, or an open legal dispute with an insurer relating to back pain.

An fMRI substudy enrols 40 of the 220 participants. Its analysis is exploratory and is not covered by this plan.

2.3 Interventions

Participants in both arms receive 12 one-to-one sessions with a qualified physiotherapist over 16 weeks. The first two sessions may last an hour and the remaining ten about 30 minutes, weekly at first and then every two weeks. The intervention is the RESOLVE programme (pain science education, graded sensory retraining, and graded motor relearning and loading), delivered by therapists who have completed the five-day training. The comparator is usual physiotherapy, delivered by therapists who have not.

Usual physiotherapy is expected to be heterogeneous between practices. What control practices actually deliver will be recorded and described, because in a pragmatic trial the comparator is itself part of the finding.

2.4 Randomisation and allocation

Practices are allocated 1:1 by the central trial coordinator at ZHAW, before the intervention therapists are trained. The allocation took place live, by drawing balls of two colours from a bag, one colour per arm, which fixed the complete allocation at once, before therapist training and before any patient was recruited. Concealment of an ongoing allocation sequence therefore does not arise. The corresponding selection risk sits at patient recruitment (section 2.5).

Two points follow for the analysis. First, the allocation used neither stratification nor covariate constraints, so no design variables need to be included in the model [6, 7]. Second, if practices were recruited in waves and later waves were allocated non-randomly, wave and treatment arm would be confounded. Any newly recruited practices will therefore be randomised as a batch [8, 9], which makes the wave a stratum of the randomisation, and variables that structure the randomisation belong in the analysis [6]. Recruitment wave will be recorded for every practice and, if practices entered in distinct waves, entered in the model as a covariate.

Participants cannot be blinded, since the intervention includes a substantial home training component. The three physical assessments (two-point discrimination, the point-to-point test and lumbar movement control) are video-recorded and rated by a blinded assessor. The remaining outcomes are self-reported and completed at home through REDCap, which removes interviewer influence. The trial statistician is not blinded.

2.5 Recruitment of participants after cluster allocation

Because practices are randomised before their patients are recruited, and because the recruiting therapists know which arm their practice is in, patient selection can differ between arms. This identification or recruitment bias is the characteristic threat to cluster randomised trials [4, 10]. We will therefore report, by arm and by practice, the number of patients screened, found eligible, approached, and consenting. Visible baseline imbalance is not by itself evidence of bias. The expected difference between arms is zero under randomisation, but with 13 practices the chance variation around zero is wide, so some imbalance is to be anticipated. Imbalance is, however, also the main signal that recruitment differed between arms, and it will be read alongside the screening counts, not tested for "significance" [4, 11].

Selection can also operate at the practice level. The 20 practices were randomised 10 to 10, and the withdrawals since then have been unbalanced, six from the intervention arm and one from the control arm, plausibly because the intervention asks more of a practice than usual care does. The practices that remain in the intervention arm are therefore a self-selected subset. We will report the withdrawals with their timing and stated reasons in the flow diagram, describe the remaining practices at both levels of the baseline table, and read the arm comparison with this asymmetry in mind.

2.6 Sample size

The target difference is 2 points on the Roland–Morris Disability Questionnaire (RMDQ, range 0–24), the smallest between-group difference the investigators consider clinically worthwhile in this population. The between participant standard deviation of the RMDQ at 18 weeks is taken from the Australian trial (intervention 3.6, SD 4.6; control 6.4, SD 5.8 [1]), giving a pooled SD of 5.2 points (the computations use the unrounded 5.23). With a two-sided $\alpha = 0.05$ and 80% power, an individually randomised trial would need 108 participants per arm [12, section 3.2.1]. Using the baseline measurement, with an assumed baseline-to-follow-up correlation $r = 0.6$, reduces this by the factor $1 - r^2$ to 69 per arm [13].

Beyond this individually randomised benchmark, a cluster randomised design must account for three things: the inflation from the intra-cluster correlation, the variation in practice sizes, and the degrees of freedom left for estimating the treatment effect [14]. We take them in that order. The first two enter through the design effect [15],

$$\mathrm{DE} = 1 + \left[\left(\mathrm{cv}^2\,\frac{k-1}{k} + 1\right)\bar{m} - 1\right]\rho, \tag{1}$$

where $\bar{m}$ is the mean number of analysed participants per practice, k the number of practices, ρ the intra-cluster correlation and cv the coefficient of variation of practice size. For the intra-cluster correlation, the values reported for primary care outcomes are usually small, with a median near 0.01 [16]. For patient-reported pain and disability outcomes the observed values sit at the low end of that distribution. The one published clinic-level estimate of ρ for the RMDQ is 0 (95% confidence interval 0 to 0.10), and it comes from a trial that ran in seven physiotherapy clinics per arm against usual physiotherapy [17, 18]. The UK BEAM trial, which randomised 181 general practices with the RMDQ as its primary outcome, found the cluster correlations smaller than it had projected [19]. Measures of the care process show systematically larger intra-cluster correlations than clinical outcomes [20], and the RMDQ is a clinical outcome. We therefore read 0.01 as a plausible value and the larger values as caution. For the size variation, unequal practice sizes can be ignored when cv < 0.23, but for trials randomising general practices cv is typically around 0.65 [15]. Physiotherapy practices are the closest analogue we have, so we plan on cv $= 0.65$. With the more optimistic cv $= 0.4$ the requirements below fall to 81, 104 and 127 per arm. With $\bar{m} = 15.4$, $k = 13$ and cv $= 0.65$, the design effect is 1.20 at $\rho = 0.01$, 1.61 at $\rho = 0.03$ and 2.02 at $\rho = 0.05$, so the requirement is 83, 111 and 139 participants per arm respectively. The trial therefore aims for 100 analysed participants per arm, and recruitment continues until 200 complete primary-outcome datasets are available. [OPEN: should we not recruit 220 anyway, as the registration commits? that would also give us a buffer] The recruitment target of 110 per arm includes a buffer of about 10% for loss to follow-up (Table 1). This covers an intra-cluster correlation up to about 0.02.

Table 1: Sample size per arm required to detect a 2-point difference in RMDQ at 18 weeks with 80% power and two-sided $\alpha = 0.05$, by intra-cluster correlation. Based on a pooled SD of 5.2 points, adjustment for the baseline score (assumed correlation 0.6), an average of 15.4 analysed participants per practice and a coefficient of variation of practice size of 0.65. The requirements use the normal approximation; the degrees-of-freedom correction for the number of practices follows in the text and in Table 2.

	Intra-cluster correlation			
	0	0.01	0.03	0.05
Design effect (eq. 1)	1.00	1.20	1.61	2.02
Analysed participants per arm	69	83	111	139
Recruited per arm (10% attrition, rounded up)	77	93	124	155

This calculation rests on two approximations. The baseline gain and the clustering inflation are applied as separate factors. An exact formula that combines them exists, and across plausible cluster-level baseline correlations the required sample shifts by less than 20% (section 6 of the accompanying script) [21]. And the intra-cluster correlation is assumed, not known, so the calculation is reported across a range of values [7].

The inputs do not weigh equally. Across the intra-cluster correlations considered the requirement runs from 83 to 139 per arm, an assumed baseline correlation of 0.5 instead of 0.6 turns the 83 at $\rho = 0.01$ into 98, and ignoring the size variation between practices altogether (cv $= 0$) would lower the 83 to 79. The number of practices acts separately, through the degrees of freedom, and matters most.

The third item, the degrees of freedom, is the one the design effect does not correct. The between-practice variance must be estimated from the practices themselves, so the treatment effect is judged against a t distribution with $k_1 + k_2 - 2 = 11$ degrees of freedom instead of a normal [22, section 10.3.1], and this correction belongs in the planning [23]. Van Breukelen and Candel turn it into a simple recipe: a sample size derived from the normal approximation, as in Table 1, needs two to three further practices per arm at a two-sided α of 0.05, three when fewer than eight practices per arm are planned [14]. Table 2 applies the correction exactly. Power is evaluated from the noncentral t distribution,

$$\text{power} = 1 - T_{\nu,\lambda}\Big(t_{1-\alpha/2,\,\nu}\Big) + T_{\nu,\lambda}\Big(-t_{1-\alpha/2,\,\nu}\Big)\,, \qquad \lambda = \frac{\delta}{\sqrt{\sigma^2(1-r^2)\,\mathrm{DE}\left(\frac{1}{\bar{m}k_1} + \frac{1}{\bar{m}k_2}\right)}}, \tag{2}$$

where δ is the target difference, σ, r and DE are as above, $\nu = k_1 + k_2 - 2$, and $T_{\nu,\lambda}$ is the distribution function of the noncentral t with noncentrality λ [12, section 3.2.1]. The variance under the root is the variance of the difference in arm means [22, eq. 7.12 and section 7.6.2]. With 13 practices, power falls short of the planned 80% (Table 2, whose first row is the current 4 versus 9 split). The analysis itself needs no modification for unequal arms. The repair is more practices: adding patients to the practices already enrolled has strongly diminishing returns [24], about ten practices per arm has been recommended as a general minimum for valid inference [14], and spreading the same 200 analysed participants over 7 practices per arm reaches the planned power at $\rho = 0.01$ (Table 2). The last block of that table shows where the limit lies. With the intervention arm held at 7 practices, five further control practices raise power by between one and five percentage points, depending on the intra-cluster correlation, because the arm with fewer practices dominates the variance of the comparison. Practices for the intervention arm are therefore worth more than practices for the control arm. Any new practices will be randomised as a batch, with the split chosen to move the overall allocation towards balance [25], and the recruitment wave will enter the model as described in section 2.4. [OPEN: confirm number and split of the additional practices once the team has decided]

All sample size and power computations are reproducible from the script `R/sample_size.R` that accompanies this paper.

Table 2: Power to detect a 2-point difference in RMDQ at 18 weeks, two-sided $\alpha = 0.05$, with a t reference distribution on $k_1 + k_2 - 2$ degrees of freedom. Every row holds the analysed sample at 200 participants, so the average practice size falls as practices are added. The first row is the current allocation, the middle block is balanced, and the last block holds the intervention arm at 7 practices while control practices are added.

		Intra-cluster correlation		
Practices (int./ctrl.)	Mean practice size	0.01	0.03	0.05
4 / 9	15.4	0.73	0.61	0.51
5 / 5	20.0	0.75	0.59	0.49
6 / 6	16.7	0.78	0.65	0.55
7 / 7	14.3	**0.81**	0.69	0.60
8 / 8	12.5	0.83	0.73	0.64
9 / 9	11.1	0.84	0.75	0.67
10 / 10	10.0	0.85	0.77	0.70
7 / 8	13.3	0.82	0.71	0.62
7 / 9	12.5	0.82	0.72	0.63
7 / 10	11.8	0.82	0.73	0.64
7 / 11	11.1	0.82	0.73	0.65
7 / 12	10.5	0.82	0.73	0.65

2.7 Data integrity and missing data

Patient-reported outcomes are captured electronically in REDCap, completed by participants at home through a personalised link. The eCRF holds eligibility, demographic, medical-history and physical-assessment data.

Data will be inspected for implausible values, range violations and duplicate records before the analysis begins, and the cleaned data set will be locked before any comparison between arms is made. We will report the amount and the pattern of missing outcome data by arm and by practice.

2.8 Analytic principles

- Participants will be analysed in the arm to which their practice was allocated, regardless of the treatment actually delivered (intention to treat).
- All treatment effects will be reported as point estimates with 95% confidence intervals. Two-sided tests at a nominal α of 0.05 accompany the primary and secondary analyses, but the interpretation will rest on the estimate and its interval, not on whether an arbitrary

threshold was crossed. Reading the interval only for whether it contains zero is the same dichotomy in another form [26, 27].

- Confidence intervals and p values will not be adjusted for multiplicity. There is one primary outcome at one time point. Secondary outcomes are explicitly labelled as secondary and will be interpreted as such [5].
- Every analysis accounts for clustering by practice.
- Analyses will be carried out in R [28]. Analysis code will be published with the results [OPEN: data too?].

2.9 Outcome definitions

2.9.1 Primary outcome

The primary outcome is back-specific disability at 18 weeks after the baseline measurement, measured with the Roland–Morris Disability Questionnaire, a 24-item instrument scored from 0 (no disability) to 24 [29, 30].

2.9.2 Secondary outcomes

The trial collects a large number of secondary endpoints, which we group here by the analysis they receive.

The patient-reported continuous outcomes, collected at 18, 26 and 52 weeks, are pain intensity (numeric rating scale); the Patient Specific Functional Scale; Work Productivity and Activity Impairment; the Neurophysiology of Pain Questionnaire; the Back Pain Attitudes Questionnaire; the Pain Self-Efficacy Questionnaire; the Depression Anxiety and Stress Scale; the Tampa Scale of Kinesiophobia; the EQ-5D-5L; the Insomnia Severity Index; the Fremantle Back Awareness Questionnaire; the epistemic-beliefs questionnaire; and the treatment expectation and perceived improvement items.

The physical outcomes, taken at 18 weeks only, are two-point discrimination and the point-to-point test of tactile acuity, and lumbar movement control, each rated from video by a blinded assessor.

Two measures are recorded once. The Goal Attainment Scale is set at baseline and scored with the therapist at 18 weeks, and the satisfaction item is collected at 18 weeks only. Both are reported descriptively.

Medication intake and treatment adherence are reported as counts. Adverse events are safety data and will be summarised descriptively by arm.

Cost-effectiveness (quality-adjusted life years and incremental cost-effectiveness ratios) and the fMRI substudy are reported separately and are not covered by this plan.

3 Analysis

3.1 Baseline description

We will describe the sample at two levels, as the CONSORT cluster extension recommends [4]. At the practice level (Table 3, panel A) we will report the number of practices per arm, the number of therapists, practice size, urban or rural setting [OPEN: should we report this?], and the number of participants contributed. At the participant level (panel B) we will report means and standard deviations for approximately normally distributed continuous variables, medians and interquartile ranges otherwise, and counts with percentages for categorical variables [31]. Baseline characteristics will not be tested for statistical significance between arms, and Table 3 shows no p values, avoiding the Table 1 fallacy [11, 32].

3.2 Primary outcome

3.2.1 Estimand

The quantity of interest is the *participant-average* difference in mean RMDQ at 18 weeks between the two arms: the difference that would be seen if every patient meeting the eligibility criteria of section 2.2 in these practices received one treatment rather than the other, with each patient counting equally. With unequally sized practices it may differ from the cluster-average effect, in which each practice counts equally [33]. We specify the participant-average effect as the estimand, following the consensus extension of the ICH E9(R1) estimand addendum to cluster randomised trials [34].

A mixed model with a random intercept weights practices by their inverse variance, which matches neither weighting exactly. Where cluster size is not informative this makes no difference

Table 3: Baseline characteristics (shell). Panel A describes the clusters, panel B the participants.

Characteristic	Intervention	Control	All
Panel A: practices			
Number of practices	xx	xx	xx
Therapists per practice, median (IQR)	xx (xx to xx)	xx (xx to xx)	xx (xx to xx)
Participants per practice, median (IQR)	xx (xx to xx)	xx (xx to xx)	xx (xx to xx)
Urban setting, n (%)	xx (xx)	xx (xx)	xx (xx)
Panel B: participants			
Number of participants	xx	xx	xx
Age, mean (SD)	xx (xx)	xx (xx)	xx (xx)
Female, n (%)	xx (xx)	xx (xx)	xx (xx)
Duration of current episode, median (IQR)	xx (xx to xx)	xx (xx to xx)	xx (xx to xx)
RMDQ, mean (SD)	xx (xx)	xx (xx)	xx (xx)
Pain intensity (NRS), mean (SD)	xx (xx)	xx (xx)	xx (xx)
EQ-5D-5L index, mean (SD)	xx (xx)	xx (xx)	xx (xx)
Pain Self-Efficacy Questionnaire, mean (SD)	xx (xx)	xx (xx)	xx (xx)
Tampa Scale of Kinesiophobia, mean (SD)	xx (xx)	xx (xx)	xx (xx)
In paid work, n (%)	xx (xx)	xx (xx)	xx (xx)
Highest education level, n (%)	xx (xx)	xx (xx)	xx (xx)

and the model is unbiased for the participant-average effect. Where outcomes or treatment effects vary systematically with practice size, the estimate may depart from both estimands [33]. Table 4 therefore includes three numbers. The estimate from model (3) with its confidence interval is the result of the trial, and the only one with a hypothesis test attached. Beside it stand two estimates that target the two estimands explicitly through their weighting, specified in section 3.2.3. Agreement between the three means the weighting question does not arise here. Divergence would mean that practice size is informative, and that the participant-average estimate is the number answering the question as posed. The distribution of practice sizes will be reported so that readers can judge this.

3.2.2 Primary model

Let y_{ijt} denote the RMDQ score of participant i in practice j at occasion t, where t indexes the baseline and 18-week measurements, let s_t equal 0 at baseline and 1 at 18 weeks, and let $x_j = 1$ if practice j was allocated to the intervention and 0 otherwise. The primary analysis is the linear mixed model

$$y_{ijt} = \beta_0 + \beta_1 s_t + \beta_2\, s_t x_j + u_j + w_{ij} + \varepsilon_{ijt}, \tag{3}$$

with a random intercept $u_j \sim N(0, \tau^2)$ for practice, a random intercept $w_{ij} \sim N(0, \nu^2)$ for participant within practice, and residual $\varepsilon_{ijt} \sim N(0, \sigma^2)$. The treatment effect at 18 weeks is β_2. The model deliberately contains no main effect of treatment, so the two arms share a single baseline mean. Dropping a main effect while keeping its interaction would ordinarily violate the variable inclusion principle and leave the interaction coefficient measuring something else [35]. Here the omitted term is the between-arm difference at baseline, which randomisation makes zero in expectation. One caveat is specific to cluster randomisation: patients enter after their practice has been allocated, so differential recruitment could shift the observed baseline means, and the unconstrained model is prespecified as a sensitivity analysis for that case. This constrained longitudinal formulation [36] matches the precision of adjusting the 18-week score for its baseline value, and either is a recommended way to use a baseline measurement of the outcome in a cluster randomised trial [37].

The model includes no covariates except any used to stratify or constrain the randomisation, which must be adjusted for [6, 7]. The first allocation used neither (section 2.4). Once further practices have been randomised as a second batch, the recruitment wave structures the allocation, and model (3) then contains the wave as an additional fixed effect.

The model will be fitted by restricted maximum likelihood. Confidence intervals and p values for β_2 will use the Kenward–Roger approximation to the degrees of freedom and the associated small-sample adjustment to the covariance matrix of the fixed effects [38], which is the standard remedy for the anti-conservatism of naive mixed-model inference when the number of clusters is small [23, 39].

3.2.3 Inference with a small number of clusters

Uncorrected mixed models and generalised estimating equations have inflated type I error rates when the number of clusters is small [39, 40], and degrees-of-freedom corrections restore the nominal rate at the number of practices available here [23]. Model (3) with Kenward–Roger degrees of freedom is therefore the single prespecified analysis of the primary outcome, in line with current guidance, which advises a correction of this kind below about 40 clusters [7] or below about 30 [41].

Two further estimates are reported beside the model result (Table 4), each targeting one of the two estimands by its weighting [22, 33].

For the participant-average effect we fit the 18-week score on the baseline score and the arm by ordinary least squares, so that every participant counts equally. This is the independence estimating equation that Kahan et al. [33] recommend for this estimand, and it remains unbiased when practice size is informative. Its confidence interval uses a cluster-robust variance with the bias reduction and the Satterthwaite degrees of freedom of Bell and McCaffrey [42].

For the cluster-average effect we compare the unweighted means of the practice-level changes from baseline, with an interval from the t distribution on $k_1 + k_2 - 2$ degrees of freedom [22, section 10.3.1]. Neither estimate is tested.

Table 4: Analysis of the primary outcome (shell). The mixed model provides the estimate and the hypothesis test. The other two rows estimate the two estimands explicitly and are not tested.

Analysis	Intervention mean (SD)	Control mean (SD)	Difference (95% CI)	p
Mixed model, Kenward–Roger (primary)	xx (xx)	xx (xx)	xx (xx to xx)	.xx
Participant-average (independence estimating equation)	—	—	xx (xx to xx)	—
Cluster-average (unweighted practice means)	—	—	xx (xx to xx)	—
Intra-cluster correlation (95% CI)	xx (xx to xx)			

3.2.4 Intra-cluster correlation

As recommended by the CONSORT cluster extension [4, 16], we will report the estimated intra-cluster correlation for the primary outcome, $\hat{\rho} = \hat{\tau}^2/(\hat{\tau}^2 + \hat{\nu}^2 + \hat{\sigma}^2)$, with a 95% confidence interval obtained by profile likelihood.

3.3 The later time points

The primary model (3) extends naturally to all four occasions on which the RMDQ is collected (baseline, 18, 26 and 52 weeks), with time as a categorical variable, the same random intercepts, and again no main effect of treatment. We fit this four-occasion model to obtain the effects at 26 and 52 weeks, which are secondary.

3.4 Comparative Bayesian analysis

The primary model (3) will also be fitted in a Bayesian formulation with the same structure. The informative priors come from the individual participant data of the Australian RESOLVE trial [1], which the investigators hold under a data sharing agreement, and specifically from the distributions of the RMDQ at baseline and at 18 weeks in each arm. Written out with its priors, the model is

$$
\begin{aligned}
y_{ijt} &\sim \mathrm{Normal}(\mu_{ijt},\ \sigma), \\
\mu_{ijt} &= \beta_0 + \beta_1 s_t + \beta_2\, s_t x_j + u_j + w_{ij}, \\
u_j &\sim \mathrm{Normal}(0,\ \tau), \qquad w_{ij} \sim \mathrm{Normal}(0,\ \nu), \\
\beta_0 &\sim \mathrm{Normal}(m_0,\ s_0), \qquad \beta_1 \sim \mathrm{Normal}(m_1,\ s_1), \\
\beta_2 &\sim \mathrm{Normal}(\hat{\delta},\ c \cdot \mathrm{se}(\hat{\delta})), \\
\tau &\sim \text{Half-Normal}(s_\tau), \qquad \nu \sim \text{Half-Normal}(s_\nu), \qquad \sigma \sim \text{Half-Normal}(s_\sigma).
\end{aligned}
\tag{4}
$$

The hyperparameters come from the Australian data: m_0 and s_0 describe the baseline RMDQ distribution, m_1 and s_1 the change from baseline to 18 weeks in the control arm, and $\hat{\delta}$ with its standard error the treatment effect from the analogue of model (3) fitted to those two occasions. The factor $c > 1$ widens that prior. Widening it in this way is the power prior of Ibrahim and Chen [43], in which the likelihood of the previous study is raised to a power a_0 between 0 and 1. For a normal likelihood that multiplies the precision by a_0, so $c = 1/\sqrt{a_0}$, and a_0 reads as those authors intend it, a relative precision parameter saying what fraction of the historical study is borrowed. They recommend such control where the previous and the current study differ, which here they do on three counts. The Australian comparator was a sham rather than usual physiotherapy, the setting was Australian rather than Swiss, and that trial randomised individuals rather than practices. We set $a_0 = 1/9$, so that $c = 3$ and the prior standard deviation is 1.7 points. The designs of Table 2 imply a Swiss standard error between 0.65 and 0.71, so the prior then supplies between an eighth and a seventh of the precision of the posterior. No single value of a_0 follows from the data, so the analysis is repeated with $a_0 = 1/4$ and $a_0 = 1/16$, and all three are reported. The Australian data give $m_0 = 9.5$, $m_1 = -3.5$ and $\hat{\delta} = -2.4$ with a standard error of 0.56, and participant-level and residual standard deviations of 3.7 and 3.6, so that $s_\nu = s_\sigma = 4$. We set $s_0 = s_1 = 2$, several times the corresponding standard errors, so that the Swiss data dominate those two priors. That trial randomised individuals and has no practice level, so s_τ rests on the intra-cluster correlation instead: values between 0.01 and 0.05 imply τ between 0.5 and 1.2, and $s_\tau = 2$ covers that range. The RMDQ at 18 weeks piles up near zero (Figure 1), and access to the raw data allows this skewed outcome distribution to be modelled directly instead of assumed normal. The RMDQ is a count of 24 items, and in every arm-by-occasion subgroup of the Australian data the beta-binomial distribution with size 24 fits as well as or better than each continuous family we examined, while the normal fails badly at 18 weeks (Figure 2). The Bayesian analysis will therefore be reported twice, once with the normal likelihood of (4) for direct comparability with the frequentist result, and once with the first two lines replaced by

$$y_{ijt} \sim \text{BetaBinomial}(24,\ p_{ijt},\ \theta), \qquad \text{logit}(p_{ijt}) = \beta_0 + \beta_1 s_t + \beta_2\, s_t x_j + u_j + w_{ij},$$

which respects the bounded count nature of the score. In that version the coefficients act on the logit scale, so the priors are re-derived there by fitting the same beta-binomial model to the Australian data, and the values above do not transfer. Each version is also fitted with a sceptical prior for β_2 in place of the informative one. Following Spiegelhalter et al. [44], that prior is normal with mean zero and a standard deviation set so that the probability of a benefit beyond the alternative hypothesis is 0.05. A target difference of 2 RMDQ points therefore gives a standard deviation of $2/1.645 = 1.22$ points. On the logit scale the same 2 points, measured down from the Australian control mean of 6.4 at 18 weeks, amount to a difference of 0.48, so the standard deviation there is 0.29.

With 13 practices the posterior for the between-practice variance is sensitive to its prior [45]. We will therefore report a sensitivity analysis over the between-practice variance prior and, for the treatment effect, the posterior probabilities that the benefit exceeds 0 and 2 RMDQ points

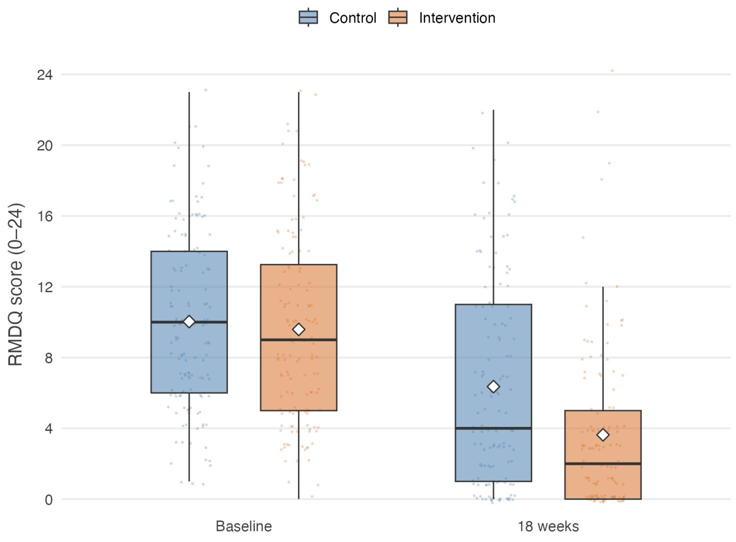

Figure 1: RMDQ in the Australian trial [1] at baseline and 18 weeks, by arm. Boxes give the interquartile range, whiskers extend to the most extreme value within 1.5 times that range, diamonds mark the means. Every participant is plotted as a point behind the boxes, scattered horizontally and vertically so that repeated integer scores remain visible.

under each prior. The Australian data will not be redistributed with this plan.

3.5 Secondary outcomes

Secondary outcomes will be analysed with models of the same structure as (3), with the link function and distribution appropriate to the outcome, always including a random intercept for practice and always using the same small-sample correction. Effects will be reported as point estimates with 95% confidence intervals (Table 5). No formal claim of efficacy will be based on a secondary outcome.

3.6 Sensitivity analyses

The following are prespecified and will be reported whether or not they change the conclusion.

1. **Leave-one-practice-out.** The primary model refitted 13 times, each time omitting one practice. With few clusters a single atypical practice can drive the result, and the spread of the 13 estimates shows how much it does.
2. **Therapist effects.** A third level for therapist within practice. This is the more faithful description of the data hierarchy, but variance components at three levels are poorly identified

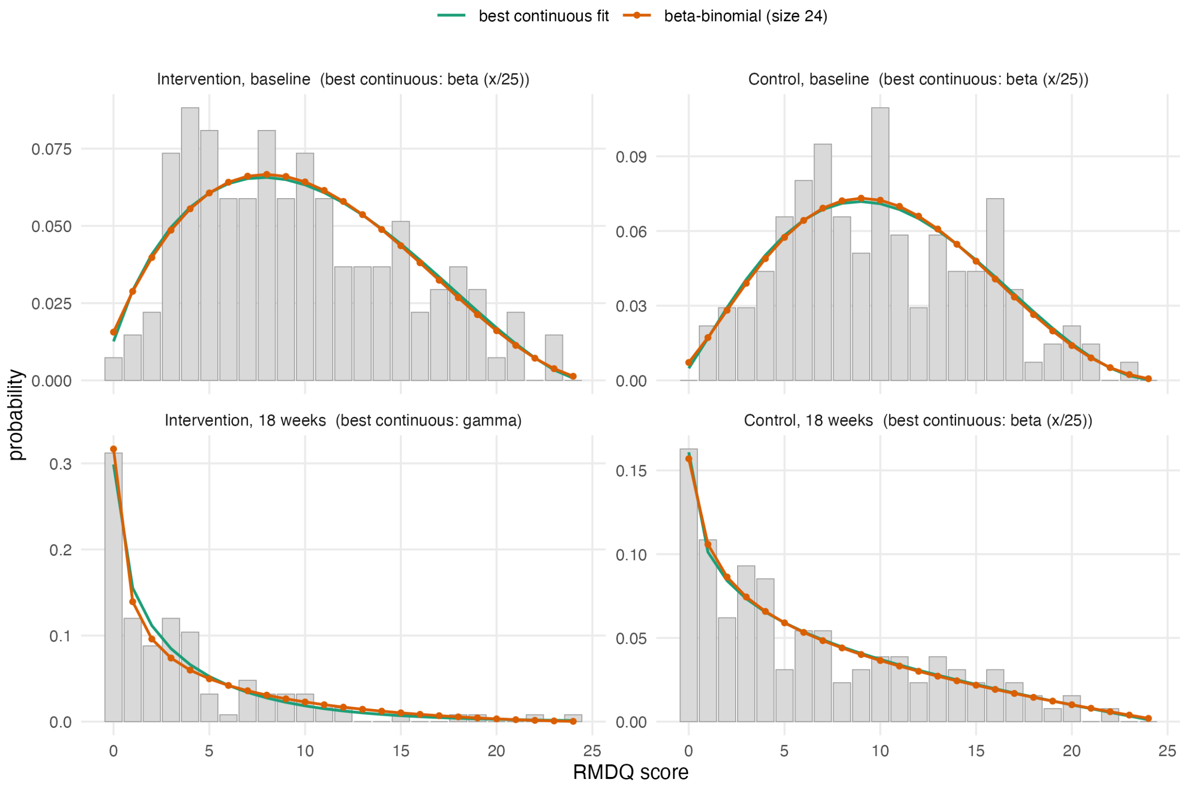

Figure 2: Observed relative frequencies of the RMDQ in the four arm-by-occasion subgroups of the Australian trial, with maximum likelihood fits evaluated on the integer grid so that the families are directly comparable. The beta-binomial with size 24 is at or near the best fit in every subgroup. Fitted values (prob, θ): intervention 0.40, 4.9 at baseline and 0.16, 3.0 at 18 weeks. Control 0.42, 6.3 at baseline and 0.26, 2.7 at 18 weeks. Script `R/aus_rmdq_distributions.R`.

with 13 practices, so it is a sensitivity analysis, not the primary model.

3. **Unconstrained model.** Model (3) with a main effect of treatment added, which frees the baseline means and shows what the constraint contributes.

4. **Cluster-mean baseline.** Model (3) with the practice mean baseline score added as a further fixed effect, which separates the within-practice from the between-practice association with baseline disability.

5. **Practice-by-time interaction.** Model (3) with an additional random practice-by-time effect, which allows the correlation between two participants of the same practice to be weaker across occasions than within one occasion. A constrained baseline analysis without this flexibility can overstate precision [37], and the time-constant practice intercept of the primary model is the less flexible variant.

Table 5: Analysis of secondary outcomes (shell).

Outcome and time point	Intervention mean (SD)	Control mean (SD)	Effect (95% CI)
Pain intensity (NRS)			
18 / 26 / 52 weeks	xx (xx)	xx (xx)	xx (xx to xx)
Patient Specific Functional Scale			
18 / 26 / 52 weeks	xx (xx)	xx (xx)	xx (xx to xx)
Tactile acuity: two-point discrimination, point-to-point			
18 weeks	xx (xx)	xx (xx)	xx (xx to xx)
Lumbar movement control			
18 weeks	xx (xx)	xx (xx)	xx (xx to xx)
Quality of life (EQ-5D-5L)			
18 / 26 / 52 weeks	xx (xx)	xx (xx)	xx (xx to xx)
Pain knowledge and attitudes (NPQ, Back-PAQ, PSEQ)			
18 / 26 / 52 weeks	xx (xx)	xx (xx)	xx (xx to xx)
Psychological measures (DASS, TSK)			
18 / 26 / 52 weeks	xx (xx)	xx (xx)	xx (xx to xx)
Sleep (ISI), body awareness (FreBAQ)			
18 / 26 / 52 weeks	xx (xx)	xx (xx)	xx (xx to xx)
Work absence (WPAI), medication intake			
18 / 26 / 52 weeks	xx (xx)	xx (xx)	xx (xx to xx)

3.7 Missing data

The primary analysis uses all participants with a baseline measurement, whether or not the 18-week outcome was recorded, and is valid under a missing-at-random assumption conditional on the observed measurements [46]. We will report the number and proportion of missing outcomes by arm and by practice, and compare the baseline characteristics of participants with and without an 18-week outcome. As a sensitivity analysis we will use multiple imputation by chained equations with the practice structure represented in the imputation model, since imputation that ignores the clustering understates the uncertainty of the imputed values [47]. If an entire practice is lost, that will be reported separately and not imputed.

3.8 Non-adherence and contamination

Adherence will be summarised as the number of the 12 sessions attended and as the proportion of participants attending at least 75% of the intervention. The mean effect in the principal stratum of participants who would adhere regardless of allocation, the complier average causal effect, will be estimated by instrumental variable regression with allocation as the instrument [48]. The estimate will be reported alongside the intention-to-treat effect, as in the Australian trial [1]. With 13 clusters this analysis is imprecise, and it is prespecified as exploratory.

The design itself limits contamination: the five-day RESOLVE training is unavailable to control therapists, and the German-language RESOLVE materials exist only inside a secured web application accessible to intervention therapists and participants. Treatment fidelity is monitored through documentation of every treatment session. We will nonetheless describe any contamination that comes to light and discuss its likely direction, since it biases the estimated effect towards the null [3].

3.9 Interim analysis

A single Bayesian interim analysis for futility is planned after 100 participants have been recruited. The trial is declared futile if the posterior probability that the between-group difference is smaller than 1 RMDQ point reaches 95%. The analysis uses model (4) with the priors of section 3.4, so the interim and final analyses are consistent. No sample size adjustment follows from the interim result.

4 Reporting

Results will be reported following CONSORT 2025 with the extension for cluster randomised trials [4, 49], including a flow diagram that tracks both practices and participants (Figure 3).

The guideline for the content of statistical analysis plans that this document follows [5] contains no items specific to clustering. An extension for cluster randomised trials is under development but not yet published [50]. We have therefore added the cluster-specific items ourselves: the estimand and its weighting, the small-sample correction, the intra-cluster correlation, and the two-level description of the sample.

Declarations

Funding. The RESOLVE Swiss trial is funded by the Swiss National Science Foundation (grant number 220585). The funder has no role in the design of the analysis or the decision to publish this plan.

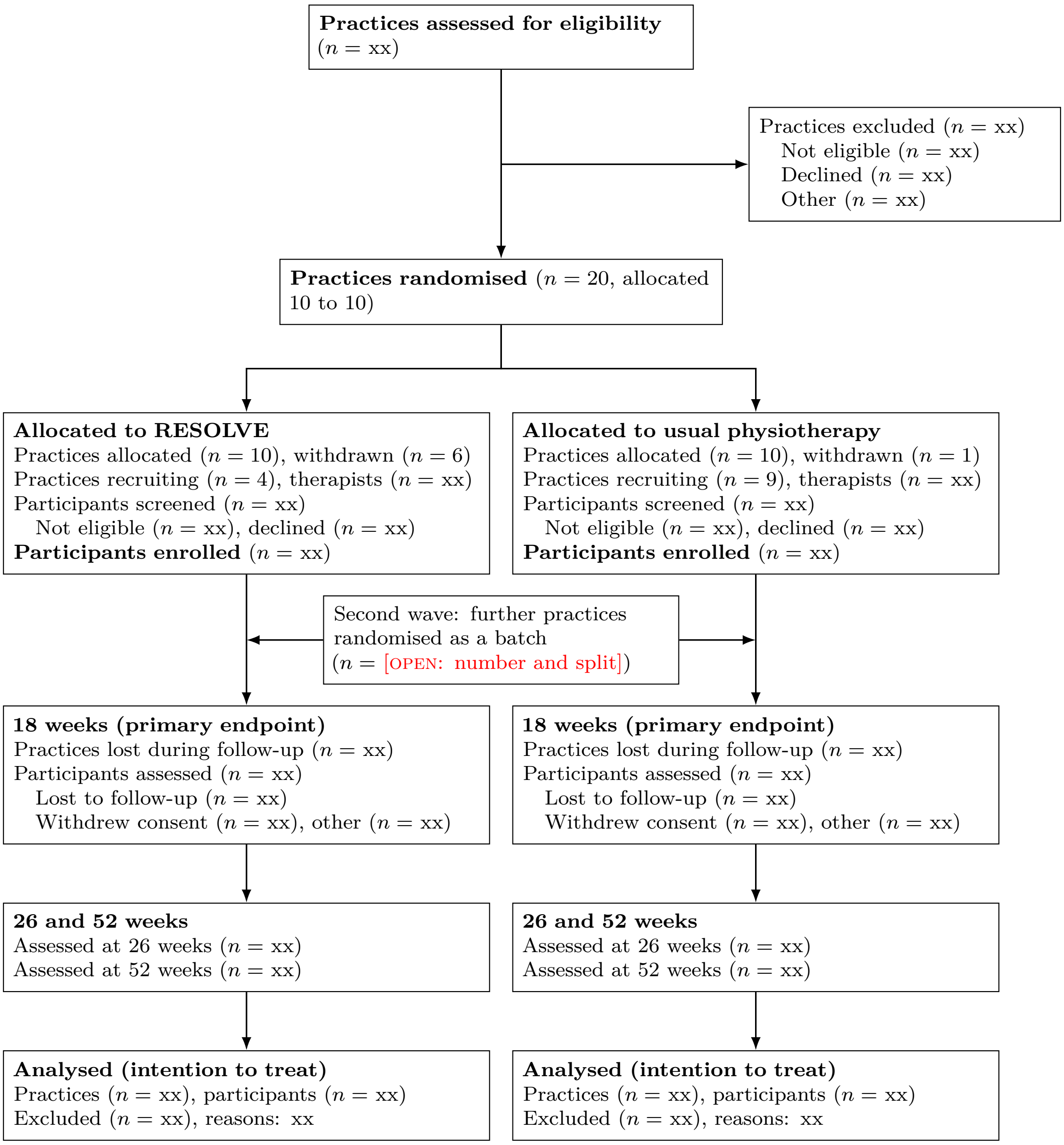

Figure 3: Flow of practices and participants through the trial (shell). Both practices and participants are counted at every stage, as the CONSORT extension for cluster randomised trials requires. The screening counts in the allocation tier are what allow recruitment bias to be judged, since practices know their allocation before they approach patients.

Conflicts of interest. [OPEN: to be completed by all authors]

Ethics. Ethical approval was granted by the Cantonal Ethics Committee of Zurich as lead committee (BASEC 2025-00784) on 20 January 2026, with conditions whose fulfilment was confirmed on 10 April 2026.

Data and code availability. The code that reproduces the sample size and power calculations reported here accompanies this paper at `https://github.com/jdegenfellner/Resolve_Swiss_Statistical_Analysis_Plan`, and the version of record will be archived with a DOI on Zenodo [OPEN: make the repository public and mint the Zenodo DOI at submission]. Analysis code for the trial will be published together with the trial results. [OPEN: may we place the trial data online, or at least the de-identified dataset behind the primary results? check the consent wording, the ethics approval and the sponsor's position]

Author contributions. JD conceived this analysis plan, developed the statistical methodology, wrote the analysis and simulation code, produced the figures, and drafted and revised the manuscript. [OPEN: add the contributions of the further authors once the author list is settled]

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